

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js workflow** for MCA filings with session handling, CAPTCHA handoff, form retries, and state recovery, reducing manual filing time by **85%**.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized backend services with **Docker**, wired **GitHub Actions** deployments to **Google Cloud Run**, and improved release velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built medical-image segmentation tooling for **95 brain subparts**, handling model inference, orientation, preprocessing, and large-volume data paths with reproducible outputs.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Worked in a mature open-source codebase with review constraints, compatibility concerns, and careful integration into existing user workflows.

Open Source

[Contributions](#) 

- Contributed patches across Invesalio, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside existing maintainer security, reliability, and review expectations.

Projects

Murdoc | *Python, FastAPI, MCP, OpenTelemetry, Prometheus, Docker*

GitHub 

- Built a self-hosted **AI security gateway** for LLM, HTTP tool, and MCP traffic, blocking **90%+** of prompt-injection and unsafe-tool attempts across a **150-case attack corpus** with false positives under **5%**.
- Added route profiles, RBAC runtime settings, PII redaction, allow/block rules, **OpenTelemetry** metrics/traces, health checks, and audit ledgers for **100%** of policy decisions.

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, AWS, Prometheus*

GitHub 

- Built containerized Go/Rod services for a **content distribution platform** using browser automation, with async workers, RabbitMQ retries, Redis state, and PostgreSQL persistence.
- Designed the system for operational simplicity, service isolation, and recoverable execution around unreliable external integrations and long-running jobs.

Distributed Inference Engine | *Python, PyTorch, FastAPI, gRPC, CUDA, Docker, Redis*

- Built a **distributed inference engine** to run models larger than a single consumer GPU by coordinating token generation across heterogeneous nodes with speculative decoding.
- Designed streaming APIs, worker heartbeats, retry-aware orchestration, and KV-cache-aware routing to balance latency, throughput, and worker availability across local and remote inference workers.

Technical Skills

Languages: Go, Python, TypeScript

Frameworks: FastAPI, Gin, Echo, Node.js

Tools: MCP, OpenTelemetry, Prometheus, PostgreSQL, Redis, Docker, GitHub Actions